TUTOR GUIDE • LEVEL 1 • TERM 1 • LESSON 5 OF 10

Make-a-Ten Adding



Big idea: Make tens first. To add, fill up a ten first. Move some from the other number to make a full ten, then add what is left.

90 minutes • live online or face to face • up to 8 children • ages 6-8 • Notice it. Test it. Explain why.

1. At a glance

Strand	Addition and Subtraction (Adding more than two 1-digit numbers; Addition within 100 by making a ten (mental method first); Written algorithms are not taught in this lesson) Strand label for the tutor only. It is not a claim that the program covers the school syllabus, and it must not appear on child or parent pages as 'MOE-aligned'. Syllabus check for the owner or a tutor: the P1 outline names adding more than two 1-digit numbers, addition within 100 and mental calculation of a 2-digit number plus ones without renaming. Crossing a ten (28 + 6) by making a ten is this program's own mental method, placed in Start here by the plan; confirm that placement.
Objectives	• Say how many more make 10 from any number to 9, and how many more make the next ten from a two-digit number. • Add two numbers that cross a ten (8 + 6, 28 + 6) by filling up a ten first and adding what is left, and explain why the total does not change. • Add three or more numbers by finding pairs that make 10 or a tens number, in any order. • Choose which number to fill up and see that moving objects between groups keeps the total the same.
Joining cold	**Remember warm-up covers:** How many more make 10 (bonds to 10) with a ten-frame and fingers; How many more make the next ten (bonds to 20: 14 + 6, and 18 + 2, 26 + 4); Tens and ones put together (30 and 6 is 36). Only counting to 20, showing a number with fingers and knowing that ten ones make a ten. Nothing from the earlier Term 1 lessons is needed: the warm-up builds tens and ones with a ten-frame and the kit. Lesson 5 is one of the hardest to miss in Term 1 (plan, section 2): Lesson 6 builds on it. A child who misses it watches the recording and does workbook pages 1 and 2 before Lesson 6.
Child materials	• About 20 small things of two kinds (buttons, pasta, clips) • Paper to draw a ten-frame on, or the tens bar from the kit page • The printed workbook and the kit page, cut out by a grown-up • A pencil, a rubber and spare paper
Tutor materials	• A slide deck and a paper ten-frame with two kinds of counters to show on camera if you wish • A visible clock No ruler is needed. Nothing needs to be bought: everything can be found at home. No matches, toothpicks or anything sharp. Small things never go near the mouth or ears.
Tiers	Start here (7 questions, pages 1 and 2), Think harder (4, page 3), Go further (3, page 4), Investigation and look-back (page 5), kit page (page 6). Labels are neutral; nobody is told which page they are on.
Contest-style move	Used at Think harder only: add by making tens (pair the numbers that make 10); reorder an addition to find pairs that make a ten or a tens number. Never advertised to children or parents.

Before the lesson

- Solve every workbook question yourself first, making a ten each time.
- Tell parents a week before: print workbook pages 1 to 6, cut out the kit page (page 6), find about 20 small things of two kinds and a sheet of paper for a ten-frame.
- Have the slides open and check the kit page is visible on screen share. Have a paper ten-frame and two kinds of counters to show on camera if you wish.
- Check the recording settings exclude children's video.

Safeguarding and recording (this lesson)

- Small things stay on the table: nothing goes in the mouth or ears. No matches, toothpicks or anything sharp.
- This lesson uses the main room only: no breakout rooms, so the tutor sees every camera. A parent sits nearby but does not give answers.
- Sneak-peek and exit answers are shown on the child's own paper to the camera at the same moment, once. There is no private chat and no board held up in front of the group.
- Never read out a child's name with a right or wrong answer. Never show a child's page in the recording; record the tutor and the screen only, with children's video excluded.
- A child who ticks 'I am still stuck' gets a check-in in the open session next lesson (a quick visit to the shared room or a hand signal, parent nearby). No private messages to children.
- The look-back is never collected in front of the group or read out. A pass is always allowed.

2. Timing plan (90 minutes exactly)

Minutes	Part	Min	Slides	Script fills
0–2	**Opening**	2	1: Make-a-Ten Adding	1:26
2–8	**Remember warm-up**	6	3: Warm-up: fill the ten-frame, How far to the next ten?, Tens and ones together	4:39
8–12	**Sneak-peek puzzles**	4	1: Sneak-peek puzzles	3:30
12–28	**Explore**	16	4: Explore: fill the ten-frame, Your turn: make a ten, Build 28 + 6 with the kit, What did we notice?	12:42
28–38	**Mini-lesson**	10	7: Fill the ten first, Split the 6, Add to a tens number: 28 + 6, Which number do we fill?, Your turn: add, Check: add, Spot the pair: 3 + 8 + 7	7:47
38–42	**Brain break**	4	1: Stomp to ten	3:12
42–70	**Workbook**	28	5: Workbook page 1: Start here, Check page 1, Workbook page 2: Start here, Check page 2, Choose your next page	25:44
70–76	**Talk time**	6	2: Talk and explain, Back to the sneak-peek	5:15
76–82	**Exit ticket**	6	2: Exit ticket, Check the exit ticket	4:14
82–85	**Look-back**	3	1: Look-back	2:28
85–90	**Flex time and goodbye**	5	1: Well done, inquirers!	0:36
	Total	**90**		**71:34**

The script needs 71:34 of the 90:00 when children are quick. A slow class (speech 10% slower, children and slide changes 20% slower) needs 84:57. The last 5 minutes are a standing buffer: if you are behind, use them; do not start anything new. Clock checks: minute 12 (end of sneak-peek), minute 42 (start of workbook), minute 70 (talk time). If more than 2 minutes behind at minute 42, skip in this order: optional items and extra quick questions first, then the second explore round, then any quick-fire in the mini-lesson, then the optional slide. Never skip the look-back or the camera and privacy moments.

3. Lesson script

Times show when each line should start, counted from the start of the lesson, if you keep to the script. SAY lines are spoken in your own words. REC lines are the short recording-voice cues for children who watch later. Children answer aloud, with fingers or hands; no private chat; no boards held up in front of the group.

0:00 – 2:00 Opening

script 1:26 of 2:00

0:00	SLIDE	Show slide **Make-a-Ten Adding** (s-title)
0:25	REC	*If you are watching the recording: hello! Get a pencil, paper, your workbook, your kit pieces and a few small things. Pause or rewind any time.*
0:38	SAY	Good morning, inquirers! Cameras on if you like, pencil ready, a few small things and your kit pieces on the table. Today we learn a trick for adding without counting one by one: make a ten first.
0:56	DO	Greet each child by first name and check their camera shows their page and hands. [0:30]
	NOTE	Remind the group once, in one line, that the lesson is recorded (you and the screen only) and that a child may keep the camera off. The routine and the four inquirer rules were introduced in Lesson 1: do not re-teach them.

2:00 – 8:00 Remember warm-up

script 4:39 of 6:00

2:00	SLIDE	Show slide **Warm-up: fill the ten-frame** (s-rem1)
2:25	REC	*Recording: join in! Show your answers on your fingers before I say them. Pause if you need longer.*
2:34	SAY	Fingers and a ten-frame. A ten-frame has ten spaces. When it is full, that is a ten. I say how many counters are in the frame. You show me, on your fingers, how many more fill it up.
2:53	ASK	Seven counters. How many more to make ten? [wait about 12 s for answers]
3:09	ASK	Four? [wait about 12 s for answers]
3:22	ASK	Nine? [wait about 12 s for answers]
3:34	ASK	Five? [wait about 12 s for answers]
3:46	ASK	Two? [wait about 12 s for answers]
3:59	SLIDE	Show slide **How far to the next ten?** (s-rem2)
4:24	SAY	Now bigger numbers. Eighteen is one ten and eight ones. How many more ones make the next ten?
4:33	ASK	Eighteen. How many more to make twenty? [wait about 12 s for answers]
4:48	ASK	Fourteen, to make twenty? [wait about 12 s for answers]
5:02	ASK	Twenty-six, to make thirty? [wait about 12 s for answers]
5:16	ASK	Thirty-nine, to make forty? [wait about 12 s for answers]
5:30	SLIDE	Show slide **Tens and ones together** (s-rem3)
5:56	SAY	Last one. Tens and ones put together. Say the tens first, then the ones.
6:02	ASK	Thirty and six more makes? [wait about 10 s for answers]
6:15	ASK	Forty and two more? [wait about 10 s for answers]
6:27	ASK	Fifty and nine more? [wait about 10 s for answers]
	IF STUCK	If a child hesitates on how many more make ten, count the empty spaces in the ten-frame slowly with them. For 26 to 30, show 26 as 2 bars and 6 squares on the kit and count the squares needed to fill the next bar. Do not move on until most children have answered; this idea is the lesson.
	NOTE	Answers: 3, 6, 1, 5, 8 then 2, 6, 4, 1 then 36, 42, 59. Children answer with fingers, hands or a whispered number. Nobody writes. Joining cold: all you need is counting to 20 and ten ones making a ten; the pictures do the rest.

8:00 – 12:00 Sneak-peek puzzles

script 3:30 of 4:00

8:00	SLIDE	Show slide **Sneak-peek puzzles** (s-sneak)
8:25	REC	*Recording: pause now and try both puzzles on your paper. Guessing is fine. Press play for the next part.*
8:34	SAY	Two sneak-peek puzzles: ideas we have not learnt yet. A guess is a great start, and I do not know is welcome. Nobody is marked.
8:47	READ	Puzzle one. Count the counters. How many are there altogether? You may draw rings round groups.
8:56	CHILDREN	Children count the counters and write the number on their own paper. [0:45]
9:41	READ	Puzzle two. Ben has 9 red counters and 5 blue counters. He moves 1 blue counter to the red counters. Now he has 10 red counters and 4 blue counters. Is the total still the same? Circle Yes or No. Then write how you know.
10:05	CHILDREN	Children write or draw their answer. [0:55]
11:00	HOLD UP	Everyone holds their paper to the camera at the same moment, once, after puzzle two. Do not read answers aloud. No private chat.
11:25	SAY	Thank you. We will come back to puzzle two later.
	NOTE	Puzzle 1 answer: 12 counters. Puzzle 2 answer: yes, nothing was added or taken away; 9 + 5 and 10 + 4 are both 14. Look for good ideas to praise later (a child who said we only moved one counter). Do not write scores or keep a list.

12:00 – 28:00 Explore

script 12:42 of 16:00

12:00	SLIDE	Show slide **Explore: fill the ten-frame** (s-explore1)
12:25	REC	*Recording: pause now. Draw a ten-frame, move your things to make a ten, then say what is left. Press play when you have tried.*
12:37	SAY	Today your hands are the teachers. Draw a ten-frame on your paper: a box with two rows of five spaces. Or lay a tens bar from your kit in front of you. A tens bar has ten spaces too.
12:56	SAY	Put 8 small things on your frame. Put 6 of another kind beside it.
13:04	ASK	How many empty spaces are left in the frame? [wait about 15 s for answers]
13:23	SAY	Now fill the frame. Move some things across from the six. Count what is left outside.
13:31	CHILDREN	Children move 2 things into the frame and count 4 left outside. [0:45]
14:16	ASK	Tell me: how many did you move, and how many are left outside? [wait about 30 s for answers]
14:52	SAY	So eight and six is the same as ten and four. Say it with me: eight and six is ten and four. Ten and four is fourteen.
15:06	SLIDE	Show slide **Your turn: make a ten** (s-explore2)
15:31	SAY	Round two. I call a sum. You build it on your frame and make a ten. Tell me how many you moved, and ten and what.
15:44	SAY	Nine and four.
15:46	CHILDREN	Children build 9 + 4 and fill the frame: move 1, ten and 3. [0:40]
16:26	SAY	Seven and five.
16:27	CHILDREN	Build 7 + 5: move 3, ten and 2. [0:40]
17:07	SAY	Six and eight.
17:08	CHILDREN	Build 6 + 8: fill the 8, move 2, ten and 4. [0:45]
17:54	SAY	Nine and seven.
17:55	CHILDREN	Build 9 + 7: move 1, ten and 6. [0:40]
18:35	ASK	Which number did you fill up each time? Why that one? [wait about 40 s for answers]

19:20	SLIDE	Show slide **Build 28 + 6 with the kit** (s-explore3)
19:46	SAY	Round three. Bigger numbers. Build twenty-eight with your kit: 2 tens bars and 8 ones squares. Put 6 more squares beside it.
19:56	ASK	How many more squares fill the 8 up to another ten? [wait about 20 s for answers]
20:22	CHILDREN	Children move 2 squares to make a third bar: 3 bars and 4 squares. [0:45]
21:07	ASK	What number is that? [wait about 12 s for answers]
21:21	SAY	Now thirty-seven and five.
21:23	CHILDREN	Build 37 + 5: move 3 squares, 4 bars and 2 squares. [0:50]
22:13	SAY	One more. Forty-six and seven.
22:16	CHILDREN	Build 46 + 7: move 4 squares, 5 bars and 3 squares. [0:50]
23:06	SLIDE	Show slide **What did we notice?** (s-explore4)
23:30	ASK	What did you notice? Which number did we fill up? How many did we move? Why do we want a ten? [wait about 50 s for answers]
24:31	SAY	A ten is easy to count. When we make a ten first, what is left is small. That is the whole trick.
	IF STUCK	A child who adds all of the second number after the ten (8 + 6 = 10 + 6): ask how many were in the 6 before, and how many did you move out. Count the counters left outside the frame.
	IF STUCK	A child who has too few things: use kit squares for both groups, or draw dots in a ten-frame.
	NOTE	Main room only: no breakout rooms in this lesson, so you can see every camera. Pairs are optional: a grown-up may check the building. Building with the ten-frame and kit is for every child; the what-if card is for any child who wants it, and nobody is named. Answers: 13, 12, 14, 16, then 34, 42, 53.

28:00 – 38:00 Mini-lesson

script 7:47 of 10:00

28:00	SLIDE	Show slide **Fill the ten first** (s-m1)
28:25	REC	*Recording: keep watching and copy the examples. Rewind if you miss a step.*
28:32	SAY	Look. Eight and six. We want a ten. The eight is nearly full.
28:38	ASK	How many more does eight need to make ten? [wait about 12 s for answers]
28:54	SLIDE	Show slide **Split the 6** (s-m2)
29:20	SAY	So we split the six into two and four. Eight and two make ten. Then ten and four is fourteen. We moved two counters, and none were lost.
29:34	ASK	Why is the total still the same? [wait about 25 s for answers]
30:02	SLIDE	Show slide **Add to a tens number: 28 + 6** (s-m3)
30:27	SAY	Now twenty-eight and six. Twenty-eight needs two more to make thirty. Split the six into two and four. Twenty-eight and two is thirty. Thirty and four is thirty-four.
30:41	ASK	Which number did we fill up, and why? [wait about 20 s for answers]
31:05	SLIDE	Show slide **Which number do we fill?** (s-m3b)
31:30	SAY	Six and eight. We could fill the frame with the six, or with the eight. Look at both pictures.
31:40	ASK	Which way moves fewer counters? [wait about 25 s for answers]
32:07	SAY	Fill the number that is nearly full. The eight only needs two. Either way, the total is fourteen.
32:16	SLIDE	Show slide **Your turn: add** (s-m4)
32:41	READ	Your turn. Add. Nine and four. Seven and six. Eighteen and five. Forty-five and seven.
32:49	CHILDREN	Children write four answers. [1:20]
34:09	SLIDE	Show slide **Check: add** (s-m4a)

34:34	**READ**	Thirteen, thirteen, twenty-three, fifty-two. Mark your own in another colour.
34:40	**SLIDE**	Show slide **Spot the pair: 3 + 8 + 7** (s-m5)
35:05	**SAY**	Last idea. Three, eight and seven. Look for two numbers that make ten. They do not have to be next to each other.
35:16	**ASK**	Which two make ten? [wait about 20 s for answers]
35:38	**SAY**	Three and seven make ten. Ten and eight is eighteen. The order does not change the total.
	NOTE	Key words to use every time: ten-frame, make a ten, split, move. Do not teach the column (written) method today: the program teaches the mental method first. Do not teach carrying.
	IF STUCK	If a child fills the wrong number or moves too many, ask: which number is nearly full? How many empty spaces does it have? Then count the empty spaces together.

38:00 – 42:00 Brain break

script 3:12 of 4:00

38:00	**SLIDE**	Show slide **Stomp to ten** (s-break)
38:25	**REC**	*Recording: stand up and join in! Stomp the number that makes ten.*
38:31	**SAY**	Stand up and get a sip of water. We stomp the number that makes ten. I say seven, you stomp three times.
38:42	**CHILDREN**	Call 7, 4, 9, 2, 6, 8. Children stomp (or tiptoe or tap) the number that makes ten and say it aloud together. [2:30]

42:00 – 70:00 Workbook

script 25:44 of 28:00

42:00	**SLIDE**	Show slide **Workbook page 1: Start here** (s-ws1)
42:25	**REC**	*Recording: pause now and do workbook page 1. When you finish a page, press play to check it.*
42:34	**READ**	Page 1, Start here. Question 1: fill the first ten-frame and add 9 and 4. Question 2: draw 7 dots and 6 dots, ring ten, and write the sums. Question 3: how many more make the next ten?
42:55	**CHILDREN**	Page 1, questions 1 to 3. Tutor walks the cameras, asks how do you know. Wait 10 seconds before helping. [8:00]
50:55	**SLIDE**	Show slide **Check page 1** (s-ws1ans)
51:20	**READ**	Question 1: one, three, thirteen. Question 2: three, thirteen. Question 3: three, two, four, one. Mark your own page in another colour.
51:32	**SAY**	After question 3 you may choose: carry on, or try Think harder. Everyone may choose, at any time.
51:41	**SLIDE**	Show slide **Workbook page 2: Start here** (s-ws2)
52:06	**READ**	Page 2. Question 4: add 6 to 28. Question 5: add 7 to 35. Question 6: four sums. Question 7: Siti adds 8 and 6. Is she right or wrong? Circle one. Then find her mistake in the second step.
52:28	**CHILDREN**	Page 2, questions 4 to 7. [9:00]
61:28	**SLIDE**	Show slide **Check page 2** (s-ws2ans)
61:53	**READ**	Two, four, thirty-four. Five, forty-two. Fifteen, twenty-three, fifty-three, sixty-four. Question 7: Siti should write fourteen, because she forgot that two of the six were already used.
62:07	**SLIDE**	Show slide **Choose your next page** (s-choice)
62:32	**SAY**	When you have finished Start here, you can choose Think harder, Go further or the Investigation. Everyone may choose. You can swap at any time.
62:44	**CHILDREN**	Choice time: Think harder (page 3), Go further (page 4), the Investigation (page 5), or finish and check your own work. Tutor never says who is on which page. [5:00]
	IF STUCK	Q1: count the empty spaces in the first frame. Q2: draw the 7 dots in a row, then 6 more, and ring the 7 and 3 more. Q4 to Q6: ask how many more make the next ten, then how many are left of the second number. Q7: ask how many were in the 6 before and how many Siti moved.

	IF STUCK	A child who is still stuck ticks 'I am still stuck' on the look-back page. You check in during the open session in the next lesson (a quick visit to the shared room or a hand signal, parent nearby). Never a private message.
	NOTE	Think harder is for the make-a-ten idea in a new setting (three or four numbers). Go further is for mastery and extension. Do not announce who is on which page. Pairs are never split unless both children agree.

70:00 – 76:00 Talk time

script 5:15 of 6:00

70:00	SLIDE	Show slide **Talk and explain** (s-talk)
70:25	REC	*Recording: pause. Tell a grown-up who is right, and show them with counters.*
70:32	READ	Mei and Ali both add eight and six. Mei moves two counters from the six to the eight. Ali moves four counters from the eight to the six. Is Mei right? Is Ali right? Use counters to show.
70:52	CHILDREN	Build it on your ten-frame both ways. How many does each way move? [0:50]
71:42	CHILDREN	Talk to the grown-up next to you or to your screen for one minute. Use the sentence starter: I think... because... [0:50]
72:32	ASK	Who would like to share? Volunteers first. A pass is always fine. [wait about 90 s for answers]
74:08	SLIDE	Show slide **Back to the sneak-peek** (s-sneakans)
74:33	SAY	Remember puzzle two? Ben moved one counter from the blue to the red. Is the total the same? Yes. Moving counters does not change how many there are.
74:47	ASK	Did anyone change their mind about puzzle two while we worked? You can just nod. [wait about 20 s for answers]
	NOTE	Model answer: both are right. Mei gets 10 and 4, Ali gets 4 and 10, and both are 14 because nothing is added or lost. Mei moved fewer, so her way is quicker. Volunteers first; never call a child by name to answer; no tallies, no rankings. Praise the reason, not the speed.

76:00 – 82:00 Exit ticket

script 4:14 of 6:00

76:00	SLIDE	Show slide **Exit ticket** (s-exit)
76:25	REC	*Recording: pause and do the exit ticket on your own. Then press play to check it.*
76:33	SAY	Exit ticket. Three questions, on your own, without looking back. Write your answers on page 5.
76:41	CHILDREN	Children answer E1 to E3 on their own. [2:30]
79:11	HOLD UP	Everyone holds their ticket to the camera at the same moment so only you see it. No private chat.
79:36	SLIDE	Show slide **Check the exit ticket** (s-exitans)
80:01	READ	Thirteen. Forty-three. Nineteen, and I added six and four first.
80:06	SAY	Check your own ticket in a different colour. Well done. I will stamp your inquiry journal.
	NOTE	Mark S, D or H in the tutor tracker (tutor only): S = secure on all three, D = developing (one wrong), H = needs help (two or three wrong). If D or H, plan two minutes of catch-up at the start of Lesson 6, in the open session, no names. Do not read out any child's result.

82:00 – 85:00 Look-back

script 2:28 of 3:00

82:00	SLIDE	Show slide **Look-back** (s-look)
82:25	REC	*Recording: pause and fill in your look-back box. You can pass.*
82:30	SAY	Look-back. Tick what was hard and what helped you. Write or draw one thing you did well. If you like, tick one thing to try next time. You can pass. Nobody reads it out.
82:48	CHILDREN	Children fill in the look-back box on page 5. [1:40]

85:00 – 90:00 Flex time and goodbye

script 0:36 of 5:00

85:00	**SLIDE**	Show slide **Well done, inquirers!** (s-end)
85:25	**SAY**	Well done, inquirers. Next lesson is Tens Jump, Ones Hop. You need a pencil and paper. Goodbye, and thank you to the grown-ups.
	SLIDE	Show slide **Go further: what if?** (s-opt) *(optional)*
	SAY	Optional, only if you are on time: what if both numbers are nine? Add nine and nine. How many do you move to make a ten?
	CHILDREN	Short think time, then take two answers. [1:30]
	SLIDE	Show slide **If we have time: pick one** (s-menu) *(optional)*
	SAY	Spare-time menu. If the whole class is ahead, pick one or two activities from the guide (about 4 minutes each).
	NOTE	This is the standing buffer. If you are behind, finish the look-back and say goodbye. Do not start anything new. Optional slide answer: 18 (move 1, ten and 8). If the whole class is ahead, use the Spare-time menu (guide section 3b) instead of rushing on.

3b. Spare-time menu: if the class is ahead

- Whole class ahead: use the Spare-time menu in the Flex time and the optional Go further slide. Pick one or two activities; do not start new content.
- A few children ahead: they may choose Go further or the Investigation, which are open to everyone. Never label or name a child, and never announce who is ahead.
- Never teach anything the next lesson needs. If the class is ahead, slow down: ask 'How do you know?' and let children explain, rather than rushing on.

Optional time, never added to the 90-minute sum. It comes from the 5-minute flex buffer, from slack in the parts, or from the optional Go further slide.

Activity	Minutes	You say	Recording cue	Answer
Is it always true?	4 (optional)	Mei says: if the ones of two numbers make 10, the total is a tens number. Try 24 + 16, 38 + 52 and 45 + 25. Then try 26 + 36. What is different?	*Recording: pause. Try 24 + 16, 38 + 52 and 45 + 25. Is the total a tens number?*	Yes when the ones make exactly 10: 24 + 16 = 40, 38 + 52 = 90, 45 + 25 = 70, because the ones fill one more bar and none are left. 26 + 36 = 62 is different: the ones are 6 and 6, which make 12, not 10, so the total is not a tens number.
Make a puzzle for your grown-up	4 (optional)	Write a sum with four numbers. Make two of them a pair that makes ten. Give it to your grown-up to solve. Can they find your pair?	*Recording: pause. Write a sum of four numbers with a pair that makes ten for a grown-up to solve.*	No single answer. Check that two numbers make 10 and that the total is right, for example 6 + 7 + 4 + 5 = 22.
Teach it back	3 (optional)	Pretend your grown-up or a toy has never seen a ten-frame. Teach them how to add 8 + 5 by making a ten.	*Recording: pause. Teach a grown-up how to add 8 + 5 with a ten-frame.*	8 needs 2 more to make 10. Move 2 from the 5, so 3 are left. 10 + 3 = 13.

4. Questions, answers, hints and common mistakes

Sneak-peek puzzles (minutes 8 to 12)

Puzzle 1 (access): Count the counters. How many are there altogether? You may draw rings round groups.

Answer: 12

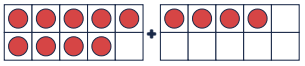
Puzzle 2 (think): Ben has 9 red counters and 5 blue counters. He moves 1 blue counter to the red counters. Now he has 10 red counters and 4 blue counters. Is the total still the same? Circle Yes or No. Then write how you know.

Answer: Yes. Nothing was added or taken away. 9 + 5 = 14 and 10 + 4 = 14.

Unmarked. Papers are shown to the camera once, together. Reveal the answers where your script says (usually Talk time).

Start here (workbook pages 1 and 2)

S1 **Look at the ten-frames. Fill in the blanks.**



a) The first frame needs ___ more counters to make 10.
b) Move them. Now there are 10 and ___ counters.
c) 9 + 4 = ___

Answer: 1 more; 10 and 3; 9 + 4 = 13

Hint (say this first): Count the empty space in the first frame. Move that many counters across.
Worked solution: The first frame has 1 empty space, so 9 needs 1 more. Move 1 counter from the 4. The first frame is full (10) and 3 are left. 10 + 3 = 13.
Common mistakes: Writing 10 and 4: forgetting to take the moved counter away from the 4.; Counting all the counters again and slipping (12 or 14).
Time for a middle child: about 2 min.

S2 **Draw 7 dots and 6 dots in the box. Ring 10 dots. Then fill in the blanks.**

7 + 6 = 10 + ___
7 + 6 = ___

Answer: 7 + 6 = 10 + 3 = 13 (ten dots ringed, 3 left over)

Hint (say this first): Draw 7 dots, then 6 dots. Ring the 7 and 3 of the other dots. How many dots are not ringed?
Worked solution: Ring the 7 dots and 3 of the 6 dots to make ten. The other 3 dots are outside the ring. So 7 + 6 = 10 + 3 = 13.
Common mistakes: Ringing ten dots but forgetting the dots left outside.; Miscounting when counting every dot one by one.
Time for a middle child: about 3 min.

S3 **Fill in the missing numbers to make the next ten.**

a) 7 + ___ = 10
b) 18 + ___ = 20
c) 26 + ___ = 30
d) 39 + ___ = 40

Answer: 3, 2, 4, 1

Hint (say this first): Look at the ones digit. How many more ones make a full ten?
Worked solution: (a) 7 and 3 make 10. (b) 18 is 1 ten and 8 ones, and 8 and 2 make 10, so 18 + 2 = 20. (c) 6 and 4 make 10, so 26 + 4 = 30. (d) 9 and 1 make 10, so 39 + 1 = 40.
Common mistakes: Writing the next ten itself (20 for part b).; Counting from the tens digit instead of the ones digit.
Time for a middle child: about 2.5 min.

S4 **Add 28 + 6. Fill in the blanks.**



a) 28 needs ___ more to make 30.
b) 6 is 2 and ___.
c) 28 + 6 = ___

Answer: 2 more; 6 = 2 and 4; 28 + 6 = 34

Hint (say this first): Take 2 from the 6 to fill up 28 to 30. How many are left of the 6?
Worked solution: 28 needs 2 more to make 30. Split 6 into 2 and 4. 28 + 2 = 30, and 30 + 4 = 34.
Common mistakes: Writing 36: adding all 6 after reaching 30, when 2 of the 6 were already used.; Splitting 6 into 2 and 3 or another wrong pair.
Time for a middle child: about 2.5 min.

S5 **Add 35 + 7. Fill in the blanks.**

a) 35 needs ___ more to make 40.
b) 35 + 7 = ___

Answer: 5 more; 35 + 7 = 42

Hint (say this first): How many more make the next ten after 35? Take that many from the 7.
Worked solution: 35 needs 5 more to make 40. 7 is 5 and 2. 35 + 5 = 40, and 40 + 2 = 42.
Common mistakes: Writing 47 (adding 7 to the 4 tens).; Forgetting the 2 left over after the ten.
Time for a middle child: about 2 min.

S6 **Add.**

a) 9 + 6 = ___
b) 18 + 5 = ___
c) 47 + 6 = ___
d) 56 + 8 = ___

Answer: 15, 23, 53, 64

Hint (say this first): Fill in the missing numbers to make the next ten. Take that many from the second number. Then add what is left.
Worked solution: (a) 9 + 1 = 10, 10 + 5 = 15. (b) 18 + 2 = 20, 20 + 3 = 23. (c) 47 + 3 = 50, 50 + 3 = 53. (d) 56 + 4 = 60, 60 + 4 = 64.
Common mistakes: Moving the right amount to make the ten but then adding the whole second number.; Making a ten with the wrong number (for example 9 + 6 = 9 + 4 + 2).
Time for a middle child: about 3 min.

S7 **Siti adds 8 + 6. Here is her working: 8 + 2 = 10, then 10 + 6 = 16. Fill in a to c.**

a) Siti's answer: right or wrong? Circle Right or Wrong.
b) The right answer is ___.
c) In her second step, 10 + 6 should be 10 + ___.

Answer: No. The right answer is 14. 10 + 6 should be 10 + 4: she moved 2 from the 6, so only 4 are left.

Hint (say this first): Siti moved 2 from the 6 to the 8. How many are left in the 6?
Worked solution: 8 + 2 = 10 uses 2 of the 6. Only 4 are left, so the sum is 10 + 4 = 14. Siti added all 6 again.
Common mistakes: Saying she is right because 10 + 6 looks like a good sum.; Writing 14 without saying what she forgot (a spoken reason is fine).
Time for a middle child: about 2.5 min.

Think harder (page 3): the taught idea in a new setting

T1 **Mei adds 4 + 9 + 6. Fill in the blanks.**

a) Two numbers that make 10 are ___ and ___.
b) The total is ___.

Answer: 4 and 6 make 10. The total is 19.

Hint (say this first): Look for two numbers that join to make 10. They do not have to be next to each other.
Worked solution: 4 and 6 make 10. 10 + 9 = 19. Check the long way: 4 + 9 = 13, and 13 + 6 = 19.
Common mistakes: Adding the first two numbers (4 + 9) and stopping or slipping.; Thinking the pair must sit side by side.
Time for a middle child: about 3 min.

T2 **Add 8 + 5 + 2 + 7. Show your working on the lines.**

The total is ___.

> **Answer: 22 (8 + 2 = 10, 5 + 7 = 12, 10 + 12 = 22)**

Hint (say this first): Find two numbers that make 10 first. Then add the other two.
Worked solution: 8 + 2 = 10. 5 + 7 = 12 (5 + 5 + 2). 10 + 12 = 22.
Common mistakes: Adding from left to right and losing track.; Using a number twice or leaving one out.
Time for a middle child: about 4 min.

T3 **Three of these sums make 30: 17 + 13, 18 + 12, 19 + 12, 16 + 14. Circle the sum that does not.**

I know because ___.

> **Answer: 19 + 12 (it makes 31). The ones 9 and 2 make 11, not 10.**

Hint (say this first): Look at the ones digits only. Which pair of ones does not make 10?
Worked solution: 7 + 3 = 10, 8 + 2 = 10 and 6 + 4 = 10, so those make 30. 9 + 2 = 11, so 19 + 12 is 31.
Common mistakes: Adding every sum one by one and slipping.; Choosing 18 + 12 because 18 looks big.
Time for a middle child: about 3 min.

T4 **Ben has 18 red marbles, 34 blue marbles and 12 green marbles. How many marbles does Ben have altogether? Show your working on the lines.**

Ben has ___ marbles altogether.

> **Answer: 64 (18 + 12 = 30, 30 + 34 = 64)**

Hint (say this first): Which two numbers make a tens number? Add those first.
Worked solution: 18 + 12 = 30. 30 + 34 = 64. The order does not change the total.
Common mistakes: Leaving out one group of marbles.; Adding 18 + 34 first and slipping on the ten.
Time for a middle child: about 4 min.

Go further (page 4): mastery and extension

G1 **Add 18 + 7 + 12 + 23 + 3 + 17. Show your working on the lines.**

The total is ___.

> **Answer: 80 (18 + 12 = 30, 7 + 3 = 10, 23 + 17 = 40)**

Hint (say this first): Which numbers have ones that make 10? Try 8 and 2, 7 and 3. Tick each number once you have used it.
Worked solution: 18 + 12 = 30. 7 + 3 = 10. 23 + 17 = 40. 30 + 10 + 40 = 80. Another way: 7 + 23 = 30, 18 + 12 = 30, 3 + 17 = 20, total 80.
Common mistakes: Adding left to right and losing the running total.; Using a number twice or leaving one out.
Time for a middle child: about 5 min.

G2 **Look at this sum: 9 + 8 + 7 + 6 + 5 + 4 + 3 + 2 + 1. Fill in a and b.**

a) The total of 1 to 9 is ___.
b) The total of 1 to 10 is ___.

> **Answer: 45 and 55 (four pairs of 10, and the 5 is left)**

Hint (say this first): Pair 1 with 9, then 2 with 8. How many pairs can you make? Which number has no partner?
Worked solution: 1 + 9, 2 + 8, 3 + 7 and 4 + 6 make four tens: 40. The 5 has no partner. 40 + 5 = 45. Adding 10 makes 55.
Common mistakes: Forgetting the 5 in the middle has no partner.; Making only three pairs.
Time for a middle child: about 6 min.

G3 **Write a sum of 6 numbers that makes 100. Many answers are right. Show your working on the lines.**

> **Answer: Any six numbers that total 100. For example 45 + 5 + 20 + 17 + 3 + 10 = 100 (45 + 5 = 50, 17 + 3 = 20, 20 + 10 = 30).**

Hint (say this first): Start with a pair that makes 50, such as 45 and 5. Then choose pairs that make 20 and 30.
Worked solution: Choose pairs first: 50 + 20 + 30 = 100. Then split each into two numbers, for example 45 + 5, 17 + 3, 20 + 10. Add them to check.
Common mistakes: Choosing numbers that total 90 or 110: add every number to check.; Using fewer than 6 numbers.
Time for a middle child: about 6 min.

Investigation (page 5): Move the fewest

Easy first step: Draw 9 dots and 6 dots. To make a ten with the 9, how many dots do you move? Now try 8 + 7.
Then: Pick two numbers from 5 to 9. You can fill the first frame or the second. Which way moves fewer dots? Does the same way always move fewer? Write how you know.

> **Model answer: Fill up the bigger number: it has the fewest empty spaces, so you move the fewest dots. The number moved is 10 minus the bigger number. 9 + 6: move 1, not 4. 8 + 7: move 2, not 3. If the two numbers are the same (7 + 7), either way moves 3.**

Look for: A child who tries several pairs, records how many dots moved each way, and notices the bigger number is closer to ten. Credit any clear reason, such as the bigger number has fewer empty spaces.
Common mistakes: Thinking you must always fill the number written first.; Not testing a pair with two equal numbers (7 + 7).

Talk and explain (minutes 70 to 76)

Mei and Ali both add 8 + 6. Mei moves 2 counters from the 6 to the 8. Ali moves 4 counters from the 8 to the 6. Is Mei right? Is Ali right? Use counters to show. Sentence starter: I think ... because ...

> **Model answer: Both are right. Mei gets 10 and 4. Ali gets 4 and 10. Both are 14, because moving counters does not change how many there are. Mei moved fewer counters (2), so her way is quicker.**

Look for: A reason about moving: nothing is added or lost, so the total stays 14. Not just both are 14. Volunteers first, a pass is always fine, never call a child by name to answer, no tallies.

Exit ticket (minutes 76 to 82): 3 questions

E1 **Add.**
7 + 6 = ___

> **Answer: 13**

Worked solution: 7 + 3 = 10. 6 is 3 and 3, so 10 + 3 = 13.
Common mistakes: Writing 16 (10 + 6).; Writing 12 or 14 after counting on fingers.

E2 **Add.**
38 + 5 = ___

> **Answer: 43**

Worked solution: 38 + 2 = 40. 5 is 2 and 3, so 40 + 3 = 43.
Common mistakes: Writing 45 (40 + 5).; Writing 33 or 83.

E3 **Add 6 + 9 + 4. Fill in the blanks.**
The total is ___.
I added ___ and ___ first.

> **Answer: 19. 6 and 4 make 10, then 10 + 9 = 19.**

Worked solution: 6 + 4 = 10. 10 + 9 = 19.
Common mistakes: Adding 6 + 9 first (15) and then slipping.; Not noticing that 6 and 4 make a ten.

Look-back (minutes 82 to 85)

Children tick or write: *Something I did well*; *The hardest part was* (knowing how many to move; splitting a number; adding to a tens number like 30 + 4; adding three or four numbers (Think harder); something else); *What helped me* (ten-frames or counters; drawing it; trying again; I asked for help; I was not stuck); *Next time I will try* (drawing it; using objects; checking my answer; asking a question); *I am still stuck (my tutor will come and help)* and *I would like to pass*. Never collected or read out in front of the group. A child who ticks still stuck gets a check-in in the open session next lesson.

5. Home practice and online practice

#	Tier	Question	Answer
H1	Start here	Fill in the blanks. 9 + 5 = 10 + ___. So 9 + 5 = ___.	4, so 9 + 5 = 14
H2	Start here	Add. 37 + 6 = ___.	43
H3	Start here	Fill in the missing numbers to make the next ten. (a) 8 + ___ = 10 (b) 45 + ___ = 50 (c) 67 + ___ = 70	2, 5, 3
H4	Think harder	Add 5 + 7 + 5 + 3. Show your working. The total is ___.	20 (5 + 5 = 10, 7 + 3 = 10)
H5	Go further	(a) Write 3 sums like 8 + 5 for a grown-up to solve. (b) Solve 3 sums that your grown-up writes for you. Many answers are right.	Any: for example 8 + 5 = 13 (fill the 8 with 2 from the 5: 10 and 3). A grown-up can check with counters.

Online practice (`out/l05/practice`, key `t1l05`): 14 items; 9 have a template that makes unlimited similar questions with a computed answer. Every template variant is checked by the validator against a brute-force oracle and the runner's own JavaScript. No timers, no scores, no streaks.

6. Parent line for the term report

Your child is learning to add by making a ten first and to say why. This week: adding within 100 by filling up a ten, using ten-frames and kit pieces.

Term line (plan, section 5): Your child is learning to see numbers as tens and ones, to notice patterns and to say why. Add one named strength and one next step from your own observation. No marks, no ranking, no claim of exam results.

7. Slide list

#	Time	Slide	On screen	Speaker note
1	0–2	**Make-a-Ten Adding** s-title	TODAY WE WILL: / Make a ten first, then add what is left.	Lesson 5 of Term 1, 90 minutes. Objective: add by making a ten first (8 + 6, 28 + 6), and find pairs that make ten in longer sums. Time plan: opening 0 to 2, warm-up 2 to 8, sneak-peek 8 to 12, explore 12 to 28, mini-lesson 28 to 38, brain break 38 to 42, workbook 42 to 70, talk 70 to 76, exit ticket 76 to 82, look-back 82 to 85, flex time and goodbye 85 to 90 (the buffer: do not start anything new). Think harder, Go further and the Investigation are for every child to choose. Nobody is told which page they should be on. Say the recording notice in one line: you and the screen only; a child may keep the camera off.
2	2–8	**Warm-up: fill the ten-frame** s-rem1	Show it on your fingers / [picture: ten_frame] / There are 7 counters in the frame. How many more make 10? / 7 4 9 5 2	A ten-frame has ten spaces. Count the empty spaces to see how many more make ten. Say each number; children show on their fingers how many more make ten. Answers: 7 needs 3, 4 needs 6, 9 needs 1, 5 needs 5, 2 needs 8. Hands up on camera is fine. No writing, no marks. A child who is new can count the empty spaces in the picture.
3	2–8	**How far to the next ten?** s-rem2	[picture: bundles 1 tens 8 ones] / 18 needs ___ more to make 20. / 18 14 26 39	A bar is a ten and each square is a one. 18 is one ten and 8 ones, so 2 more ones make another ten: 20. Then ask 14 (6 more to make 20), 26 (4 more to make 30) and 39 (1 more to make 40). Answers: 2, 6, 4, 1. These are number bonds to 20 and to the next ten. Show the kit squares filling the next bar if a child is unsure.
4	2–8	**Tens and ones together** s-rem3	[picture: bundles 3 tens 6 ones] / 30 + 6 40 + 2 50 + 9 / Say the tens first, then the ones.	Answers: 36, 42, 59. Say the tens word slowly, then the ones: thirty... six. This is the bridge to adding to a tens number: after we make a ten we say, for example, thirty and four. A child who has not met tens and ones only needs to hear and say the number words. Show 30 as three bundles of ten sticks if a child is unsure.
5	8–12	**Sneak-peek puzzles** s-sneak	Ideas we have not learnt yet. A guess is a great start. Nobody is marked. / PUZZLE 1: anyone can try: / Count the counters. / How many are there altogether? / You may draw rings round groups. / [picture: ten_frame] / PUZZLE 2: think: / Ben has 9 red counters and 5 blue counters. / He moves 1 blue counter to the red counters. / Now he has 10 red counters and 4 blue counters. / Is the total still the same? Circle Yes or No. Then write how you know.	Puzzle 1 (12 counters, 7 and 5 in two ten-frames) is an access question: any child can try it by counting or drawing rings. Puzzle 2 is a thinking question on the lesson's key idea: moving does not change the total. Say: two sneak-peek puzzles, ideas we have not learnt yet; a guess is a great start and I do not know is welcome; nobody is marked. Children write on their own paper. Everyone holds the paper to the camera once, at the same moment, after puzzle 2; there is no private chat and no board in front of the group. Answers: 12; yes (9 + 5 and 10 + 4 are both 14). Do not reveal puzzle 2 until Talk time. Do not write scores or keep a list of who found it easy.
6	12–28	**Explore: fill the ten-frame** s-explore1	Draw a ten-frame on paper: two rows of five boxes. Or lay a tens bar from your kit. / Put 8 small things on it. / Put 6 other things beside it. / Move some across to fill the frame. How many did you move? / WHAT IF?: / Fill the frame a different way. What do you notice?	Round 1, about 5 minutes. Objects first, no writing yet. Children put 8 things on the frame, 6 of another kind beside it, then move 2 across to fill the frame. They count the 4 left outside and say: eight and six is ten and four. A tens bar from the kit page is a ten-frame in a line: ten spaces. A child with few things can use kit squares of two colours (pencil dots on half of them). The what-if card previews the Investigation (filling the other frame) and is open to everyone; no child is named for choosing it.

7	12–28	**Your turn: make a ten** s-explore2	Use your ten-frame. For each sum, say how many you move, then ten and ___. / 9 + 4 7 + 5 6 + 8 9 + 7 / I move ..., so it is ten and ... / [picture: ten_frame]	Round 2, about 5 minutes, main room. Call each sum aloud and show it on the slide. Children build it on their frame and say how many they moved. Answers: 9 + 4: move 1, ten and 3 (13). 7 + 5: move 3, ten and 2 (12). 6 + 8: fill the 8, move 2, ten and 4 (14). 9 + 7: move 1, ten and 6 (16). On 6 + 8 some children fill the 6 (move 4): that is fine and works too; it is just more moves. Do not name children.
8	12–28	**Build 28 + 6 with the kit** s-explore3	[picture: bundles 2 tens 8 ones] / Put 6 more squares beside it. Move squares to fill the 8 up to a ten.	Round 3, about 5 minutes. Children build 28 with 2 tens bars and 8 ones squares, and put 6 squares beside it. Two squares fill the 8 up to a ten: that makes a third bar. Now there are 3 bars and 4 squares: 34. Then 37 + 5 (move 3: 4 bars and 2 squares, 42) and 46 + 7 (move 4: 5 bars and 3 squares, 53). Each child has 20 ones squares, enough for all of these. Everyone holds their build to the camera on a count of three, if you wish.
9	12–28	**What did we notice?** s-explore4	Which number did we fill up? Why? / How many did we move each time? / What do we say after the ten: ten and ...?	Take one or two answers from volunteers. Key ideas: we fill up the number that is nearly full; the number we move is the number of empty spaces; after the ten we say ten and whatever is left; moving does not change the total. Repeat a child's answer in your own words so the recording picks it up.
10	28–38	**Fill the ten first** s-m1	[picture: ten_frame] / 8 needs 2 more to make 10	8 + 6. The first frame has 2 empty spaces, so 8 needs 2 more. Ask: how many more does 8 need to make 10? Answer: 2.
11	28–38	**Split the 6** s-m2	[picture: ten_frame] / 8 + 6 = 8 + 2 + 4 = 10 + 4 = 14	Split 6 into 2 and 4. 8 + 2 = 10. Then 10 + 4 = 14. Ask: why is the total still the same? We only moved counters from one frame to the other; none were added and none were lost.
12	28–38	**Add to a tens number: 28 + 6** s-m3	[picture: bundles 2 tens 8 ones] / 28 needs 2 more to make 30. Split 6 into 2 and 4. / 28 + 6 = 28 + 2 + 4 = 30 + 4 = 34	Think aloud: 28 is 2 tens and 8 ones. 8 and 2 make a ten, so 28 and 2 make 30. 6 is 2 and 4. 30 and 4 more is 34. Say the tens first. Ask: which number did we fill up? Why? Answer: 28, because it is nearly at the next ten.
13	28–38	**Which number do we fill?** s-m3b	[picture: ten_frame] / Fill the 8: move 2 / [picture: ten_frame] / Fill the 6: move 4 / Fill the number that is nearly full	6 + 8 can be made into a ten two ways. Fill the frame with the 8 (it has 2 empty spaces, so move 2: ten and 4) or fill the frame with the 6 (it has 4 empty spaces, so move 4: ten and 4 again). Both give 14. The quicker way moves fewer counters: fill the number that is nearly full. This is the seed of the Investigation, so do not push the rule: ask and listen.
14	28–38	**Your turn: add** s-m4	Write on your paper. Make a ten first. / 9 + 4 7 + 6 18 + 5 45 + 7	Each child writes four answers on their own paper. Walk the cameras. Do not ask children to hold boards up in front of the group. Some children may find the answer by counting on: that is fine, ask them to try a ten as well.
15	28–38	**Check: add** s-m4a	Sum | How we made a ten / 9 + 4 | 9 + 1 = 10, 10 + 3 = 13 / 7 + 6 | 7 + 3 = 10, 10 + 3 = 13 / 18 + 5 | 18 + 2 = 20, 20 + 3 = 23 / 45 + 7 | 45 + 5 = 50, 50 + 2 = 52	Answers: 13, 13, 23, 52. Read each answer and the way to make a ten aloud so a child watching the recording can mark their own work.
16	28–38	**Spot the pair: 3 + 8 + 7** s-m5	3 8 7 / Which two numbers make 10? They do not have to be next to each other. / 3 + 7 = 10, then 10 + 8 = 18	A worked example for Think harder. In a sum with three numbers, look for two that make ten, even if they are not side by side. 3 + 7 = 10 and 10 + 8 = 18. Check: 3 + 8 = 11 and 11 + 7 = 18, the same. The order does not change the total. Keep it to about a minute and do not teach written column addition.

17	38–42	**Stomp to ten** s-break	I say a number. You stomp how many more make ten. / Try it with 7, then 4, then 9, then 2. / Stand up. Sip some water.	Four minutes. Stand up on camera. Call 7, 4, 9, 2, 6, 8 (answers 3, 6, 1, 8, 4, 2). Children stomp softly, or tiptoe or tap their fingers in a small space, and say the number aloud. No equipment.
18	42–70	**Workbook page 1: Start here** s-ws1	Questions 1 to 3. Take your time. / questions S1, S2, S3	Read questions 1 and 2 aloud. Children work on their own; the tutor walks the cameras and asks how do you know. Wait 10 seconds before helping. After question 3 any child may choose to carry on or try Think harder. Question 2: draw dots and ring ten of them.
19	42–70	**Check page 1** s-ws1ans	answers S1, S2, S3	Read each answer aloud. Children mark their own page in a different colour. Never ask a child to show a wrong answer to the group.
20	42–70	**Workbook page 2: Start here** s-ws2	Questions 4 to 7. Take your time. / questions S4, S5, S6, S7	Read question 7 aloud: it is a spot-the-mistake. Children may use their kit or ten-frame to decide. Question 6 has four sums: children make a ten for each.
21	42–70	**Check page 2** s-ws2ans	answers S4, S5, S6, S7	Read each answer aloud so a child watching the recording can mark their own work. Question 7 answer: no, Siti forgot that 2 of the 6 were already used, so it is 10 + 4 = 14. Never ask a child to show a wrong answer to the group.
22	42–70	**Choose your next page** s-choice	Page | What it is / Think harder (page 3) | Pairs that make ten in longer sums / Go further (page 4) | Bigger sums and a puzzle to write / Investigation (page 5) | Move the fewest / Check | Finish and check your own work / Everyone may choose. You can swap at any time.	Invite the whole group at once, for example: when you have finished Start here, you can choose Think harder, Go further or the Investigation. Never announce who is on which page and never name a child. Pairs are not split unless both children agree. Model answers are in the tutor guide. About 5 minutes.
23	70–76	**Talk and explain** s-talk	MEI AND ALI ADD 8 + 6: / Mei moves 2 counters from the 6 to the 8. / Ali moves 4 counters from the 8 to the 6. / Is Mei right? Is Ali right? Use counters to show. / I think ... because ...	One minute of talk to the grown-up next to you, or to your screen. Then volunteers first; a pass is always fine; never call a child by name to answer. Model answer: both are right. Mei makes 10 and 4. Ali makes 4 and 10. Both are 14 because moving counters does not change how many there are. Mei moved 2 and Ali moved 4, so Mei's way is quicker. Repeat each child's answer in your own words for the recording.
24	70–76	**Back to the sneak-peek** s-sneakans	[picture: ten_frame] / 9 + 5 / [picture: ten_frame] / 10 + 4 / Yes! Moving a counter does not change the total. Both are 14.	Reveal puzzle 2 now. Say: many of you wrote a good reason. Do not read out anyone's earlier answer. Both pictures show 14 counters: Ben moved 1 counter from the blue group to the red group, nothing was added or taken away.
25	76–82	**Exit ticket** s-exit	On your own. No looking back. / questions E1, E2, E3	Children answer on page 5. After the time, everyone holds the ticket to the camera at the same moment, so only the tutor sees it (no private chat). Stamp the inquiry journal.
26	76–82	**Check the exit ticket** s-exitans	answers E1, E2, E3	Read each answer aloud. Children check their own ticket in a different colour.

27	82–85	**Look-back** s-look	[look-back box]	Three minutes. Children tick or write the look-back box on page 5. A pass is always allowed. It is never marked, never collected in front of the group and never read out. If a child ticks I am still stuck, check in during the open session next lesson (a quick visit to the shared room or a hand signal, parent nearby); never a direct message. The Next time I will try row is optional; at the start of the next lesson you may say to the whole group: today you could try drawing it, using objects or checking your answer.
28	85–90	**Well done, inquirers!** s-end	Show a grown-up one thing you found out today. / NEXT LESSON: / Tens Jump, Ones Hop	Say goodbye and thank the grown-ups. Minutes 85 to 90 are flex time: the buffer. If you ran late, finish any open item (stamps, the look-back). Do not start anything new. If the whole class is ahead, show the optional slides and use the Spare-time menu from the tutor guide.
29	optional	**Go further: what if?** s-opt	Look at 9 + 9. How many do you move to make a ten? / [picture: ten_frame]	Optional, only if you are on time. Answer: 18. Move 1 counter from one frame to the other: 10 and 8. Take two answers and move on. Optional extension: is 9 + 9 the biggest total from two single digits?
30	optional	**If we have time: pick one** s-menu	Activity | What we do / Is it always true? | If the ones of two numbers make 10, is the total always a tens number? / Make a puzzle | Write a sum with four numbers for your grown-up, with a pair that makes ten / Teach it back | Show how to add 8 + 5 with a ten-frame / Optional. Everyone may join.	Optional Spare-time menu, used only if the whole class is ahead. Pick one or two activities (3 to 5 minutes each); full tutor lines and answers are in the tutor guide. Never start new content and never name who is ahead. Slow down with how do you know rather than rushing on. These minutes come from the flex buffer and are not part of the 90-minute sum.

8. Recording cues and chapter markers

Say each cue once, to the camera, at the start of the part. Add a chapter marker at each time. A child watching alone cannot ask you a question, so say the page and question number aloud, read each question and each answer aloud, and repeat a child's answer in your own words.

Chapter	Part	You say (recording voice)	A child watching later
0:00	Opening	*If you are watching the recording: hello! Get a pencil, paper, your workbook, your kit pieces and a few small things. Pause or rewind any time.*	Gather your workbook, pencil, paper, kit pieces and small things. Press play when ready.
2:00	Remember warm-up	*Recording: join in! Show your answers on your fingers before I say them. Pause if you need longer.*	Play along. Show each answer on your fingers before the video does.
8:00	Sneak-peek puzzles	*Recording: pause now and try both puzzles on your paper. Guessing is fine. Press play for the next part.*	Pause. Try both puzzles on your own paper. A guess is fine.
12:00	Explore	*Recording: pause now. Draw a ten-frame, move your things to make a ten, then say what is left. Press play when you have tried.*	Pause. Draw a ten-frame and make a ten with your things each time I call a sum. Write what you find.
28:00	Mini-lesson	*Recording: keep watching and copy the examples. Rewind if you miss a step.*	Watch and copy the examples. Rewind any step you missed.
38:00	Brain break	*Recording: stand up and join in! Stomp the number that makes ten.*	Stand up and stomp how many more make ten.
42:00	Workbook	*Recording: pause now and do workbook page 1. When you finish a page, press play to check it.*	Pause. Do page 1, then play to check. Then page 2. Then choose a page to try.
70:00	Talk time	*Recording: pause. Tell a grown-up who is right, and show them with counters.*	Pause. Explain your answer to someone at home, with counters and a reason.

76:00	Exit ticket	*Recording: pause and do the exit ticket on your own. Then press play to check it.*	Pause. Do the exit ticket alone, then play to check it.
82:00	Look-back	*Recording: pause and fill in your look-back box. You can pass.*	Pause. Fill in the look-back box, or pass.
85:00	Flex time and goodbye	*Recording: well done for finishing! Show a grown-up one thing you found out.*	Show a grown-up your workbook and tell them one thing you found out.

9. Timing check (model, not a stopwatch)

Part	Allocated	Script	Slow class	Flag
opening	2:00	1:26	1:41	ok
remember	6:00	4:39	5:30	ok
sneak	4:00	3:30	4:06	ok
explore	16:00	12:42	15:04	ok
mini	10:00	7:47	9:11	ok
break	4:00	3:12	3:49	ok
work	28:00	25:44	30:43	ok
talk	6:00	5:15	6:14	ok
exit	6:00	4:14	5:02	ok
look	3:00	2:28	2:55	ok
flex	5:00	0:36	0:43	ok
Total	90:00	71:34	84:57	

Assumptions as docs/audit/timing_p1.md: speech 120 words a minute, read-aloud 110, slide change 25 s, a group question 10 s thinking plus 20 s answers, papers to the camera 30 s. Workbook: Start here has 7 questions; a middle child needs about 7.5 min for page 1 and 10.0 min for page 2 (a slower child about 1.7 times as long, so may finish page 1 and part of page 2; that is fine, the check-together slides cover every answer). The lesson has no breakout rooms, so no breakout time is counted.